

Distance Meter Projects

Arduino Uno R3 + Tinkercad | HC-SR04 Ultrasonic Sensor — Basic & LCD Versions

Is document mein do projects cover kiye gaye hain: **Project 1 - Distance Meter (Basic)** jo result Serial Monitor par dikhata hai, aur **Project 2 - Distance Measurement with LCD** jo result seedha 16x2 LCD screen par show karta hai. Dono projects HC-SR04 ultrasonic sensor use karte hain. Har project ka code is PDF mein copy-paste ke liye bhi maujood hai, aur sath mein alag .ino files bhi di gayi hain jo seedha Tinkercad ya Arduino IDE mein open ki ja sakti hain.

Project 1: Distance Meter (Basic - Serial Monitor)

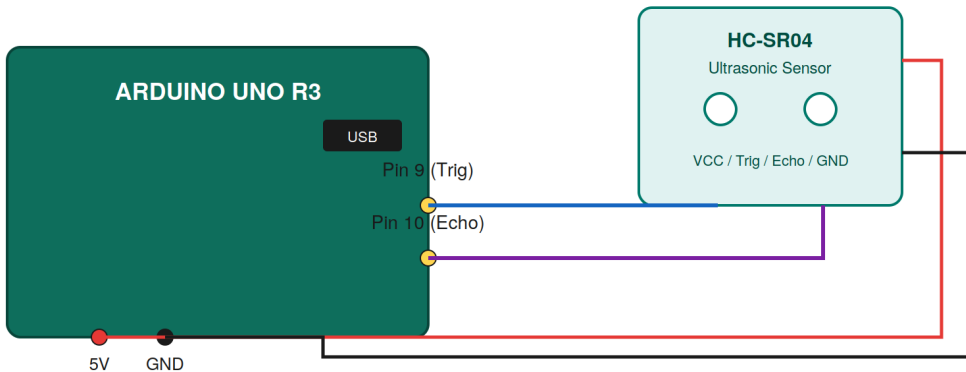
1.1 Components Used

Component	Quantity	Purpose
Arduino Uno R3	1	Main microcontroller board
HC-SR04 Ultrasonic Sensor	1	Measures distance using sound waves
Breadboard	1	For connections
Jumper Wires	As needed	For wiring all components

1.2 Circuit Diagram

White background par color-coded wiring niche di gayi hai (legend diagram ke neeche hai):

Distance Meter (Basic) — Wiring Diagram



Legend: — 5V (Power) — GND — Trig (Output) — Echo (Input)
Note: HC-SR04 VCC connects to Arduino 5V and GND connects to Arduino GND.

1.3 Build Steps

- 1 Tinkercad open karo aur naya circuit banao.
- 2 Arduino Uno R3 aur breadboard components search karke drag karo.
- 3 HC-SR04 ultrasonic sensor bhi search karke breadboard pe rakho.
- 4 Wiring karo: HC-SR04 VCC to Arduino 5V, GND to Arduino GND, Trig to Arduino Pin 9, Echo to Arduino Pin 10.
- 5 Code editor open karo aur niche diya gaya code paste karo.
- 6 Simulation start karo aur Serial Monitor open karke distance values dekho (cm mein).

1.4 Full Arduino Code

```
#define trigPin 9
#define echoPin 10

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  long duration, distance;

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(500);
}
```

Tip: Is code ko seedha copy-paste kar sakte ho, ya sath diya gaya **Distance_Meter_Basic.ino** file directly Tinkercad/Arduino IDE mein open kar lo.

1.5 Wiring Summary Table

Component Pin	Arduino Pin	Pin Type
HC-SR04 - VCC	5V	Power

Component Pin	Arduino Pin	Pin Type
HC-SR04 - GND	GND	Ground
HC-SR04 - Trig	Pin 9	Digital Output
HC-SR04 - Echo	Pin 10	Digital Input

1.6 Kaise Kaam Karta Hai

- Arduino Trig pin (9) se 10 microseconds ka HIGH pulse bhejta hai, jisse HC-SR04 se ek ultrasonic (40kHz) sound wave nikalti hai.
- Yeh sound wave saamne wali cheez se takra kar wapas aati hai, aur sensor apne Echo pin (10) ko us waqt tak HIGH rakhta hai jab tak echo wapis nahi aati.
- Arduino **pulseIn()** function se yeh maaptta hai ke Echo pin kitni der HIGH raha (microseconds mein) — yeh 'duration' hai.
- Distance nikalne ka formula: **distance = duration * 0.034 / 2**. Sound ki speed ~0.034 cm/microsecond hoti hai, aur 2 se divide isliye karte hain kyunki wave jaake wapas aati hai (round trip).
- Har 500 milliseconds baad yeh process loop() mein dobara chalta hai aur naya distance value Serial Monitor par print hoti hai.

Project 2: Distance Measurement with LCD

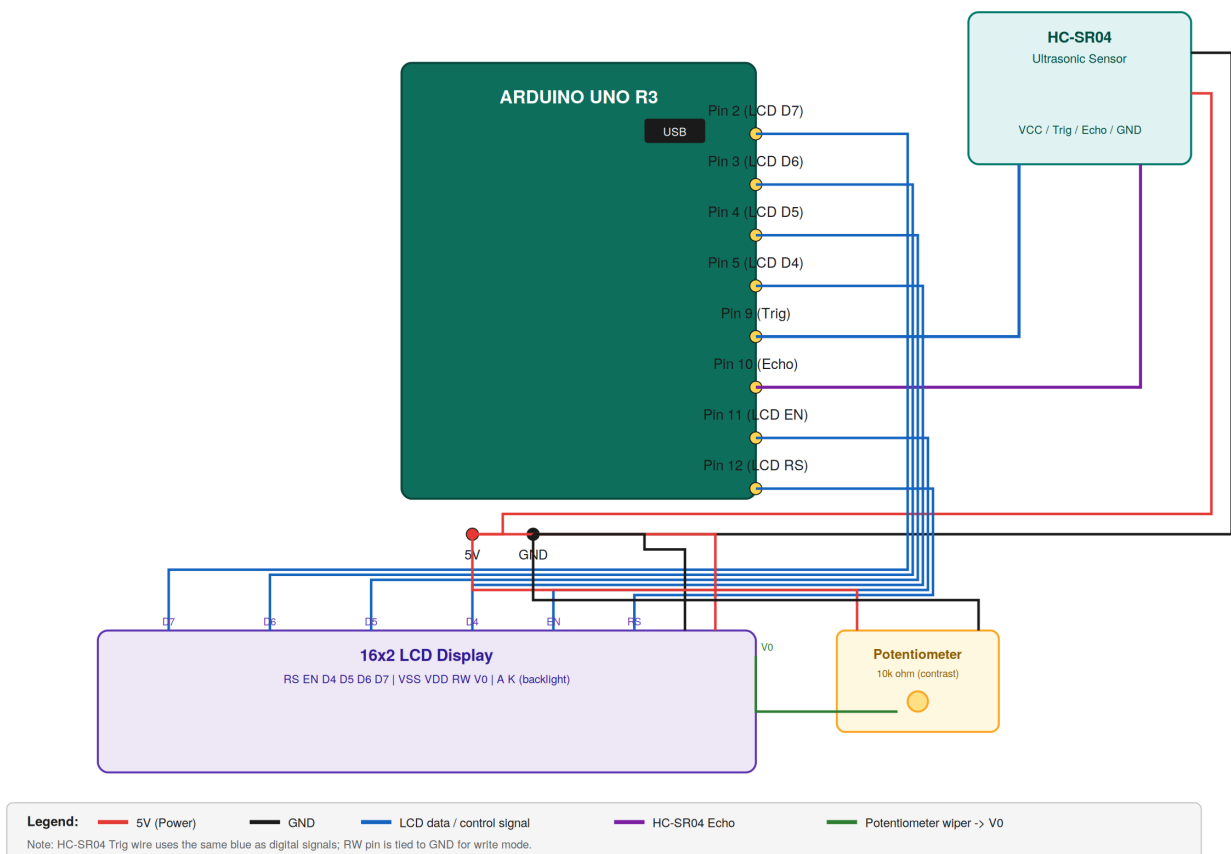
2.1 Components Used

Component	Quantity	Purpose
Arduino Uno R3	1	Main microcontroller board
HC-SR04 Ultrasonic Sensor	1	Measures distance using sound waves
16x2 LCD Display	1	Shows live distance reading
Potentiometer (10k ohm)	1	Adjusts LCD contrast
Breadboard	1	For connections
Jumper Wires	As needed	For wiring all components

2.2 Circuit Diagram

White background par color-coded wiring niche di gayi hai (legend diagram ke neeche hai):

Distance Meter with LCD — Wiring Diagram



2.3 Build Steps

- 1 Tinkercad open karo aur naya circuit banao.
- 2 Arduino Uno, breadboard, HC-SR04, 16x2 LCD, aur potentiometer sab drag karke place karo.
- 3 HC-SR04 wiring (Project 1 jaisi): VCC to 5V, GND to GND, Trig to Pin 9, Echo to Pin 10.
- 4 LCD wiring: RS to Pin 12, EN to Pin 11, D4 to Pin 5, D5 to Pin 4, D6 to Pin 3, D7 to Pin 2, VSS to GND, VDD to 5V, RW to GND, backlight A to 5V, backlight K to GND.
- 5 Potentiometer wiring: dono ends 5V aur GND se, middle pin (wiper) LCD ke V0 se.
- 6 Code editor open karo aur LiquidCrystal.h library wala code paste karo.
- 7 Simulation run karo — LCD screen par live distance dikhna chahiye.

2.4 Full Arduino Code

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

#define trigPin 9
#define echoPin 10

void setup() {
  lcd.begin(16, 2);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  lcd.print("Distance Meter");
  delay(1500);
  lcd.clear();
}

void loop() {
  long duration, distance;

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  lcd.setCursor(0, 0);
  lcd.print("Distance:");
  lcd.setCursor(0, 1);
  lcd.print(distance);
  lcd.print(" cm ");

  delay(500);
}
```

Tip: Is code ko seedha copy-paste kar sakte ho, ya sath diya gaya **Distance_Meter_LCD.ino** file directly Tinkercad/Arduino IDE mein open kar lo.

2.5 Wiring Summary Table

Component Pin	Arduino Pin	Pin Type
HC-SR04 - VCC	5V	Power
HC-SR04 - GND	GND	Ground
HC-SR04 - Trig	Pin 9	Digital Output
HC-SR04 - Echo	Pin 10	Digital Input
LCD - RS	Pin 12	Digital Output
LCD - EN	Pin 11	Digital Output
LCD - D4	Pin 5	Digital Output
LCD - D5	Pin 4	Digital Output
LCD - D6	Pin 3	Digital Output
LCD - D7	Pin 2	Digital Output
LCD - VSS / RW	GND	Ground
LCD - VDD	5V	Power
LCD - V0	Potentiometer wiper	Contrast Control
LCD - Backlight A	5V	Power
LCD - Backlight K	GND	Ground
Potentiometer - Ends	5V / GND	Power

Note: Arduino Uno ke Pin 0 aur Pin 1 serial communication ke liye reserved hain, isliye is project mein use nahi kiye gaye.

2.6 Kaise Kaam Karta Hai

- HC-SR04 wahi tarike se distance measure karta hai jaise Project 1 mein — Trig pulse bhejna, Echo pulse ka duration parhna, aur formula se cm mein convert karna.
- Farq sirf itna hai ke result Serial Monitor ki bajaye **LiquidCrystal.h** library ke zariye 16x2 LCD screen par likha jata hai.
- **lcd.begin(16, 2)** LCD ko initialize karta hai (16 columns, 2 rows). **lcd.setCursor(col, row)** batata hai ke agla text kahan likhna hai, aur **lcd.print()** asal value LCD par display karta hai.
- Potentiometer LCD ke V0 pin se juda hota hai taake screen ka contrast manually adjust kiya ja sake — agar text nazar na aaye to pot ko ghuma kar theek kiya jata hai.
- RW pin ko direct GND se joda gaya hai kyunki hum LCD ko hamesha 'write mode' mein rakhte hain (kabhi LCD se data parhna nahi hota).
- Har 500 milliseconds mein loop() naya distance measure karta hai aur LCD ki dusri row update karta rehta hai, isliye reading real-time mehsoos hoti hai.

